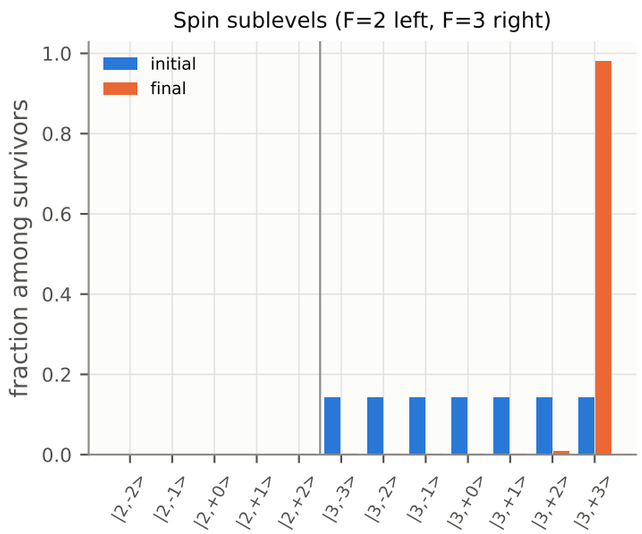
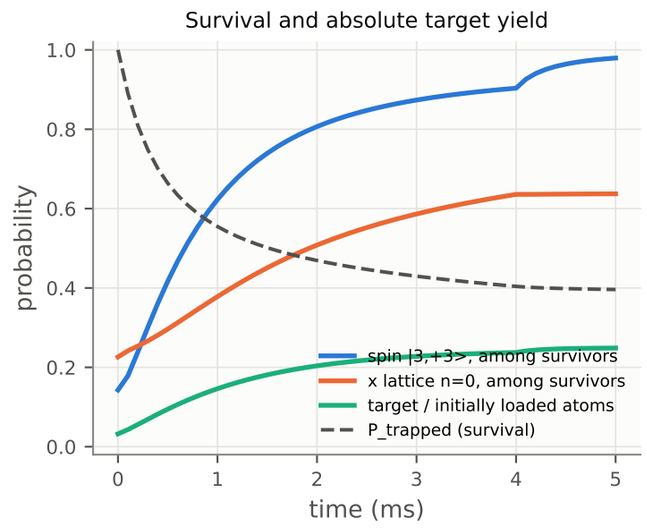
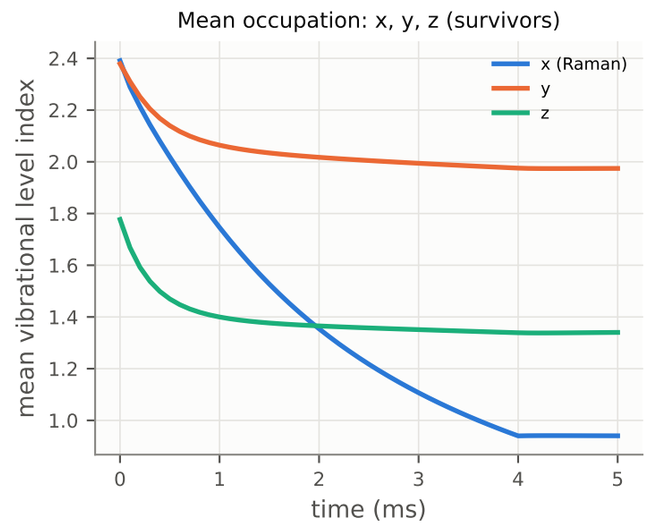
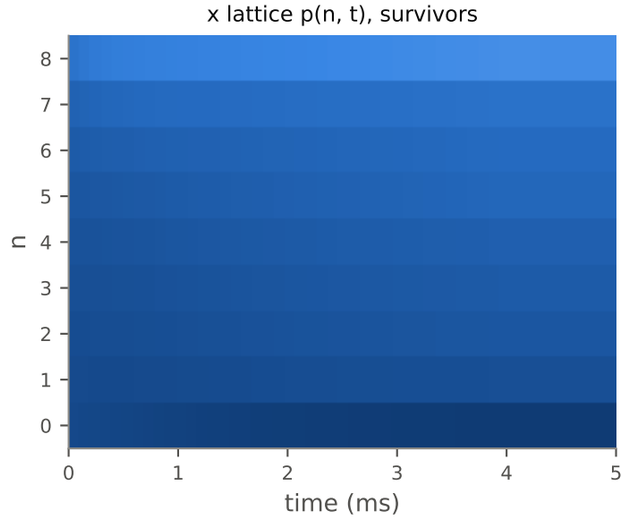
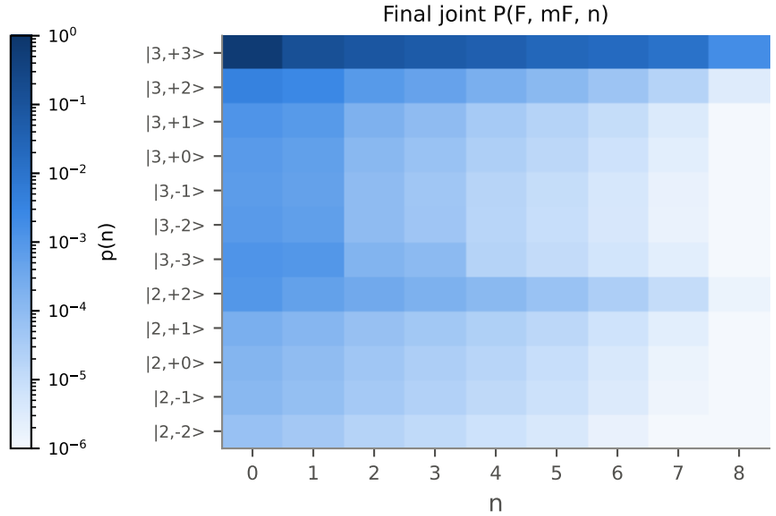


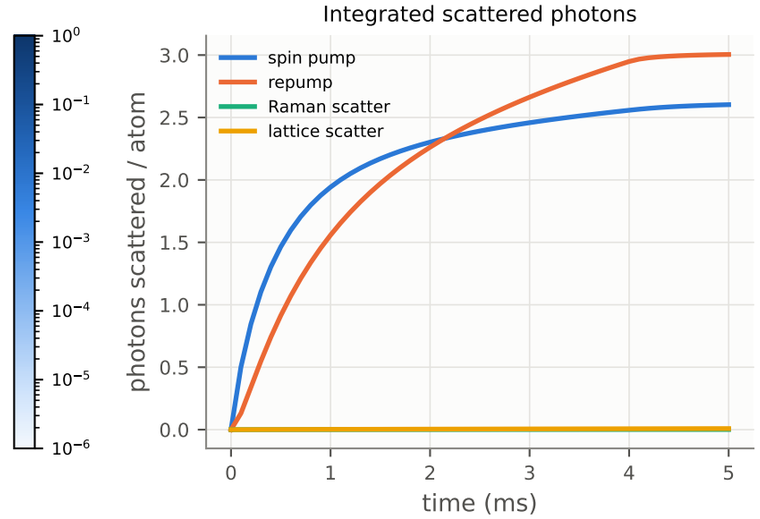


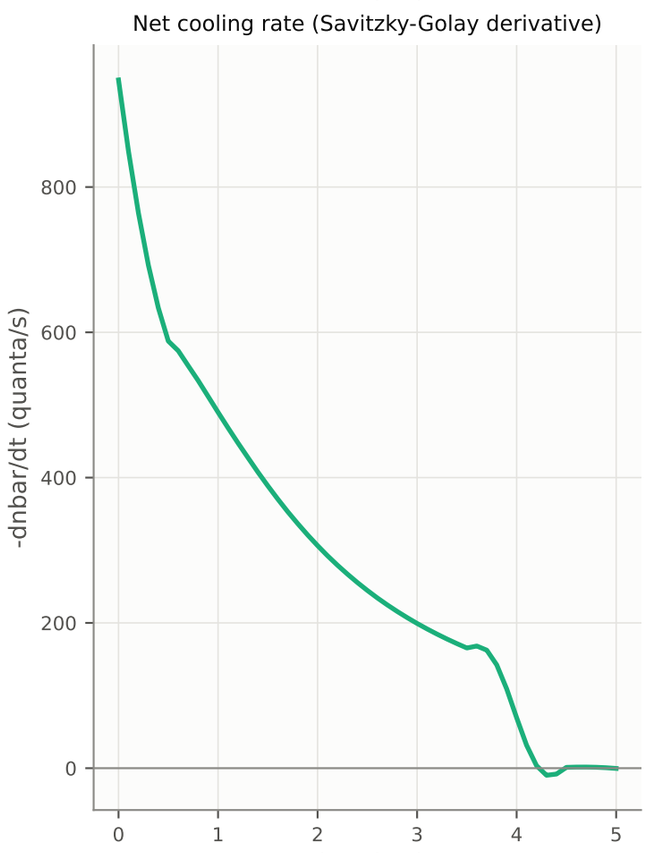













```
MODEL VALIDITY: WARNING
Raman cooling axis: local x; n and target refer to this lattice.
Raman pair: A=|3,+3> <-> B=|2,+2>; P_up and target refer to A, P_down to B.
P_up=0.9794  P_n0=0.6373  P_target=0.6276  (P_up*P_n0=0.6242)
Well-loss fraction, assuming no recapture: 0.6039
Axis temperature estimates, initial -> final (uK): x: 10 -> 2.18, y: 10 -> 6, z: 50 -> 8.29
Thermal x/y/z plots separate net energy removal from preferential-loss selection; a falling survivor temperature alone does not
establish Raman cooling.
Axis temperature proxies (uK): x=2.18, y=6, z=8.29
populations are per trapped atom: trapped 0.3961 (lost 0.6039); of all atoms: P_up=0.3880  P_n0=0.2524  P_target=0.2486
nbar (trapped atoms) 2.391 -> 0.9404   T_equiv=2.18 uK [Energy-matched mixture of local bound canonical distributions, weighted by
surviving spatial populations; proxy only. None means no finite positive-temperature match.]
photons/atom: pump 2.6, repump 3, Raman 0.00134
target |3,+3> scattering rate 27.5 /s; quanta removed per photon None
t(P_up>=0.9) = 3.84 ms;  t(P_target>=0.5) = 2.62 ms
conservation err 3.3e-14; min eig 0; boundary pop unspecified; overflow 0.6
solver coherent_3d_split_ensemble, wall 77.5 s

[OK     ] conservation: max |survival + well loss - 1| = 3.33e-14
[OK     ] positivity: minimum density eigenvalue / population = 0
[OK     ] weak excitation: maximum eliminated excited fraction across nodes and spin states = 0.0006162
[WARNING] motional secular: max optical rate / smallest local motional angular spacing across all axes = 0.301
[OK     ] carrier calibration: Modeled spatial/thermal decay; RMS carrier frequency = 5000 Hz fixes initial curvature. No fitted
homogeneous damping; reported decay is not a calibration constraint or an experimental validation.
[WARNING] three-dimensional recoil: Joint spin/nx/ny/nz populations retained, but emission uses resolved spherical dipole channels
with factorized Cartesian marginals. Interference between emitted spherical channels and correlations of one photon's Cartesian
recoil components are neglected.
[WARNING] recoil coherence: Population-resolving scattering erases motional coherence and treats elastic spin amplitudes as
distinguishable; this is an additional approximation beyond secular averaging.
[WARNING] lattice: Separable isolated wells with Gaussian local depths; no tunnelling, coherent Raman continuum excitation, envelope
transport or recapture. Scalar trap basis with diagonal differential shifts; trap shape changes from spin-dependent shifts
neglected.
[WARNING] Stark shift approximation: Raman/trap Stark operators use diagonal matrix elements; static off-diagonal motional dressing
and trap-basis changes are omitted. Excited-state lattice shifts of the pumping transitions remain the configured common optical
shift.
[WARNING] spatial ensemble: 64 frozen Gaussian position / uniform registration samples. Refine samples and time_step_us for
convergence; samples are numerical quadrature, not experimental error bars.
[WARNING] temperature: Axis temperatures are energy-matched bound-state mixture proxies; survivor cooling of undriven axes may
result from preferential loss.
```